

PROJECT REPORT

Multiclass Sleep Health Risk Classification & Explainable AI System

An End-to-End Machine Learning Pipeline for Predicting Insomnia and Sleep Apnea from Lifestyle and Physiological Health Data

Supervised Learning • Class-Imbalance Handling • Explainable AI (XAI)

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Dataset: Sleep Health & Lifestyle Records (10,000 individuals, 13 features) • 3 Target Classes: No Sleep Disorder • Insomnia • Sleep Apnea
Leakage-Free Pipeline • Stratified Cross-Validation • SHAP-Based Model Interpretability • Flask Deployment

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1. Executive Summary

This report documents an end-to-end machine learning project that classifies an individual's sleep health status using demographic, lifestyle, and physiological data. The system distinguishes between three outcomes — **No Sleep Disorder**, **Insomnia**, and **Sleep Apnea** — from a dataset of 10,000 patient records spanning occupation, activity, stress, sleep quality, and cardiovascular indicators.

The project follows a leakage-aware, ten-phase pipeline: project definition, initial data preparation, data understanding, exploratory data analysis, feature engineering, preprocessing, model development, final evaluation, probability calibration, and explainability. Five classification algorithms — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost — were each benchmarked under two class-imbalance strategies (class weighting and SMOTE) using Stratified 5-Fold Cross-Validation, since the target is imbalanced at roughly 78% / 12% / 10% across its three classes.

Headline Result

A tuned XGBoost classifier was the best-performing model, achieving **90.95% accuracy** and a **Macro F1-score of 83.41%** on a completely unseen, held-out test set — outperforming Logistic Regression, Decision Tree, Random Forest, and SVM under both imbalance-handling strategies tested. The model also reached a **balanced accuracy of 81.38%** and a **ROC-AUC of 96.77%**, confirming strong overall multiclass discrimination.

Beyond raw predictive accuracy, the project prioritised clinical interpretability. SHAP (SHapley Additive exPlanations) was applied to the final XGBoost model to quantify which features drive each prediction, confirming that **Heart Rate**, **Sleep Duration**, and **Systolic BP** are the dominant physiological and behavioural signals, while demographic and occupational attributes contribute comparatively little.

1.1 What This Project Demonstrates

- **End-to-end ML pipeline design** — from raw Excel ingestion to a deployable, explainable classifier.
- **Imbalanced-classification discipline** — stratified splitting, class weighting, and SMOTE, compared side-by-side rather than assumed.
- **Leakage-safe engineering** — preprocessing and resampling refit independently inside every cross-validation fold.
- **Feature engineering for health data** — pulse pressure and a composite sleep-quality index, each derived without touching the target.
- **Breadth across model families** — linear, tree-based, kernel-based, and gradient-boosted classifiers.
- **Explainable AI (XAI)** — global and interaction-level SHAP analysis, turning a 'black-box' gradient-boosted model into a transparent, stakeholder-readable decision system.

2. Project Overview

2.1 Project Objective

Develop an explainable multiclass classification system that predicts an individual's sleep-disorder status — No Sleep Disorder, Insomnia, or Sleep Apnea — from demographic, occupational, lifestyle, and physiological attributes, while benchmarking multiple algorithms and imbalance-handling strategies to identify the most reliable, generalisable model.

2.2 Business Objective

- Flag individuals at risk of a sleep disorder from routinely available health and lifestyle data.

- Benchmark class-imbalance strategies and select the strongest model on class-aware metrics, not raw accuracy alone.
- Explain predictions with SHAP to support clinical trust and stakeholder transparency.
- Package the final model as a deployable Flask web application for real-time inference.

2.3 Problem Statement

Sleep disorders such as Insomnia and Sleep Apnea are frequently under-diagnosed despite measurable ties to lifestyle and physiological signals — stress, activity level, blood pressure, and heart rate. Given an individual's demographic, lifestyle, and physiological attributes, this project predicts sleep-disorder status as one of three classes, using a dataset that is naturally imbalanced (78% / 12% / 10%), which makes naïve accuracy an unreliable evaluation metric and motivates the model-selection strategy used throughout this report.

2.4 Dataset Description

The dataset consists of synthetically generated patient-level sleep, lifestyle, and cardiovascular records.

Attribute	Value
Raw Records	10,182 rows (182 exact duplicates removed)
Modelling Records	10,000 records × 13 features
Target Variable	Sleep Disorder (No Sleep Disorder · Insomnia · Sleep Apnea)
Feature Groups	Demographic, Occupational, Lifestyle, Physiological, Cardiovascular
Missing Values	12 of 13 features affected (up to ~5.40% per feature)
Duplicate Records	182 identified and removed
Source	Sleep Health and Lifestyle Dataset (Kaggle, synthetically generated)

Data cleaning note: the combined “Blood Pressure” field (e.g. “126/83”) was split into separate **Systolic BP** and **Diastolic BP** columns to enable numerical analysis. The literal string “None” in the target column was explicitly preserved as a valid category during ingestion (rather than parsed as a missing value) and relabelled to “No Sleep Disorder”, and inconsistent categorical entries (e.g. “FEMALE”, “female”, “Female ”) were standardised prior to encoding.

3. Data Preparation & Understanding

Before analysis, the raw dataset was cleaned and validated to ensure every downstream statistic and model is computed on a trustworthy, consistent table.

3.1 Initial Data Preparation

- **Identifier removal:** non-predictive identifier columns (e.g. Person ID) were dropped before analysis.
- **Duplicate handling:** 182 exact duplicate records were identified and removed, leaving 10,000 unique rows.

- **Categorical standardisation:** inconsistent casing and stray whitespace in Gender, Occupation, and BMI Category were normalised to a single canonical spelling per category.
- **Missing-value standardisation:** blank strings were explicitly converted to NaN so that numerical and categorical imputers could detect them consistently.
- **Blood Pressure decomposition:** the combined "Systolic/Diastolic" text field was split into two numeric columns, Systolic BP and Diastolic BP.

3.2 Data Quality & Validation Checks

Following the initial preparation steps, the dataset was re-validated to confirm the cleaning stage produced a leakage-free, analysis-ready table.

Check	Result
Duplicate records remaining	0
Features with missing values	12 of 13 (imputed later inside the pipeline, not dropped)
Maximum missing-value rate	~5.40% (Physical Activity Level)
Target class integrity	"None" correctly preserved and relabelled — not read as null
Categorical consistency	Standardised (no duplicate-meaning categories remain)

3.3 Descriptive & Target Distribution Summary

Descriptive statistics confirmed plausible physiological ranges across the dataset (e.g. Heart Rate roughly 44–126 bpm, Sleep Duration roughly 4.5–9.4 hours), with no impossible or out-of-range values detected. The target variable, however, is meaningfully imbalanced — a fact that shapes the modelling strategy for the remainder of this report.

Class	Count	% of Dataset
No Sleep Disorder	7,813	78.13%
Insomnia	1,237	12.37%
Sleep Apnea	950	9.50%

Key takeaway: a model that always predicts "No Sleep Disorder" would already score ~78% accuracy while never identifying a single at-risk patient. This motivated **Macro F1** as the primary model-selection metric, alongside class weighting and SMOTE as imbalance-handling strategies.

4. Exploratory Data Analysis (EDA)

Before modelling, the cleaned dataset was explored visually and statistically to understand the shape of the target, feature distributions, categorical breakdowns, correlations, and outliers.

4.1 Target Variable Distribution

The target variable shows a clear three-way class imbalance: healthy individuals dominate the dataset, while Insomnia and Sleep Apnea together make up roughly one in five records.

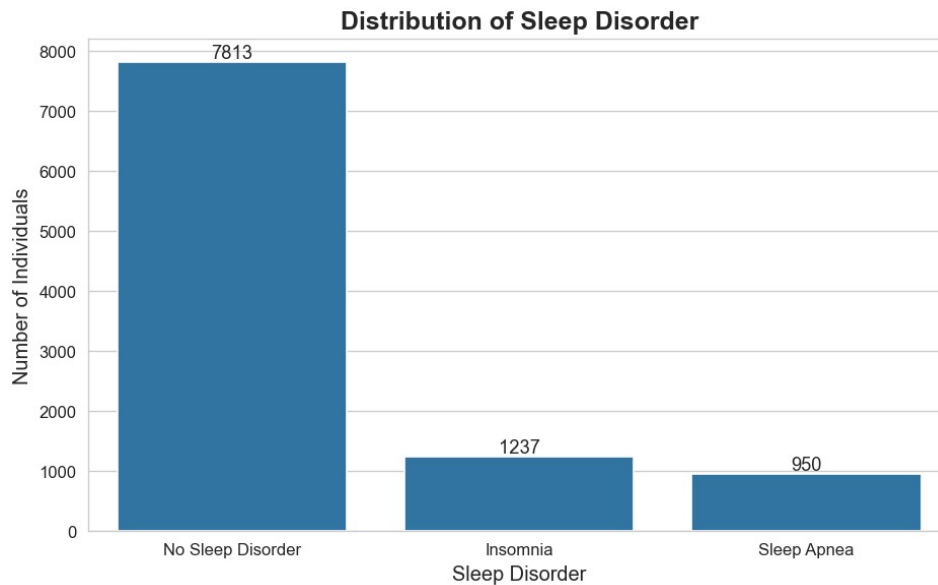


Figure 4.1 — Distribution of the Sleep Disorder Target Variable

Observation: the 78% / 12% / 10% split confirms the dataset is imbalanced rather than merely skewed, reinforcing the decision to evaluate models on Macro F1 and Balanced Accuracy rather than accuracy alone.

4.2 Numerical Feature Distribution

Histograms with kernel-density overlays were plotted for all nine original numerical features to inspect shape, skew, and multimodality ahead of modelling.

Distribution of Numerical Features

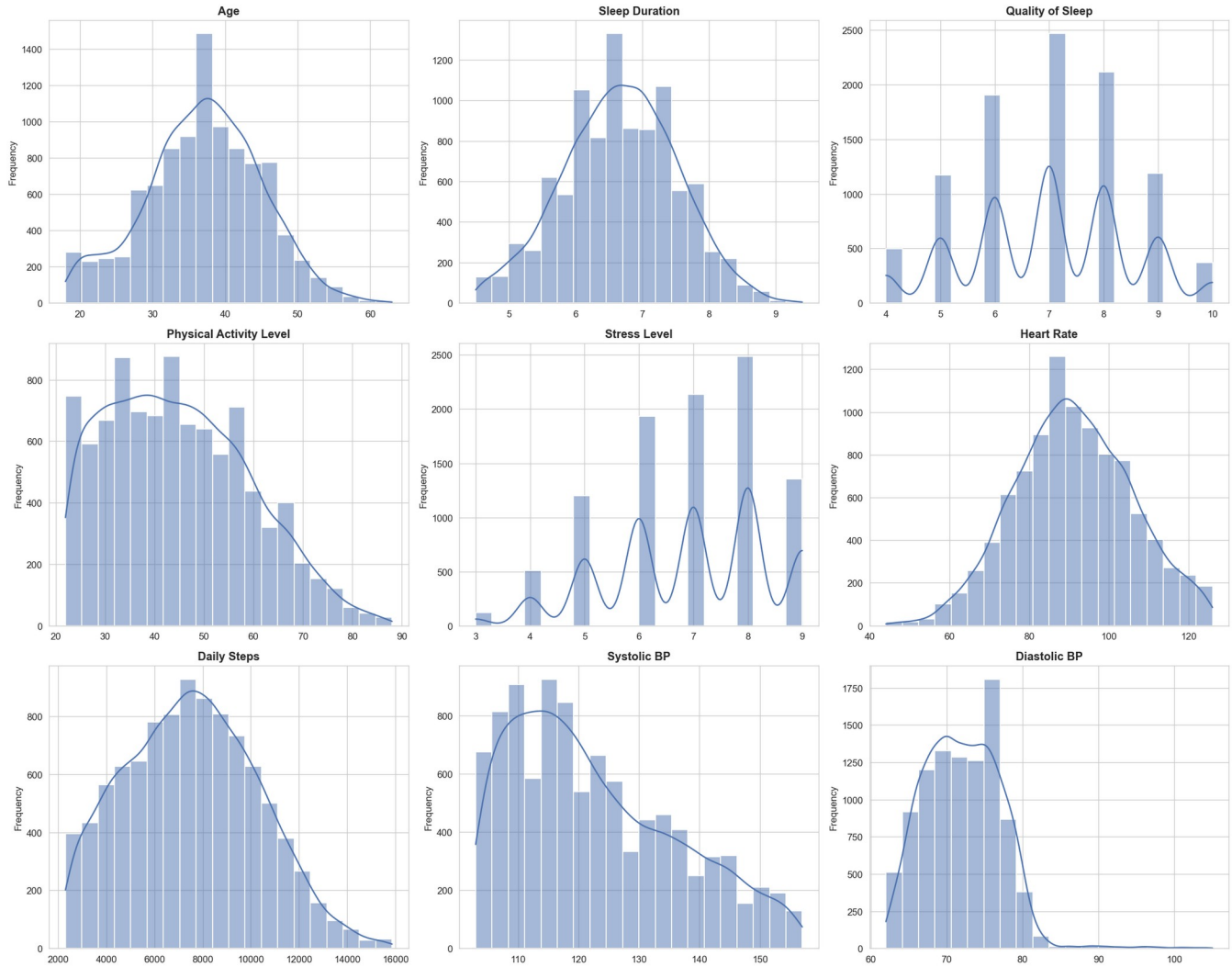


Figure 4.2 — Distribution of Numerical Features

Insights: Age is concentrated around 30–45 years with a mild right tail toward older ages; Sleep Duration is primarily distributed between 6–8 hours. Quality of Sleep and Stress Level are discrete, multi-modal ordinal scales rather than continuous measures, with observations concentrated around 7–8 and 6–8 respectively. Heart Rate is approximately bell-shaped, centred between 85–100 bpm, while Daily Steps and Systolic BP show mild right-skew consistent with a smaller, more sedentary, higher-risk subgroup. Diastolic BP is primarily concentrated between 65–80 mmHg.

4.3 Categorical Feature Distribution

Gender, Occupation, and BMI Category were profiled to understand demographic composition and category cardinality ahead of one-hot encoding.

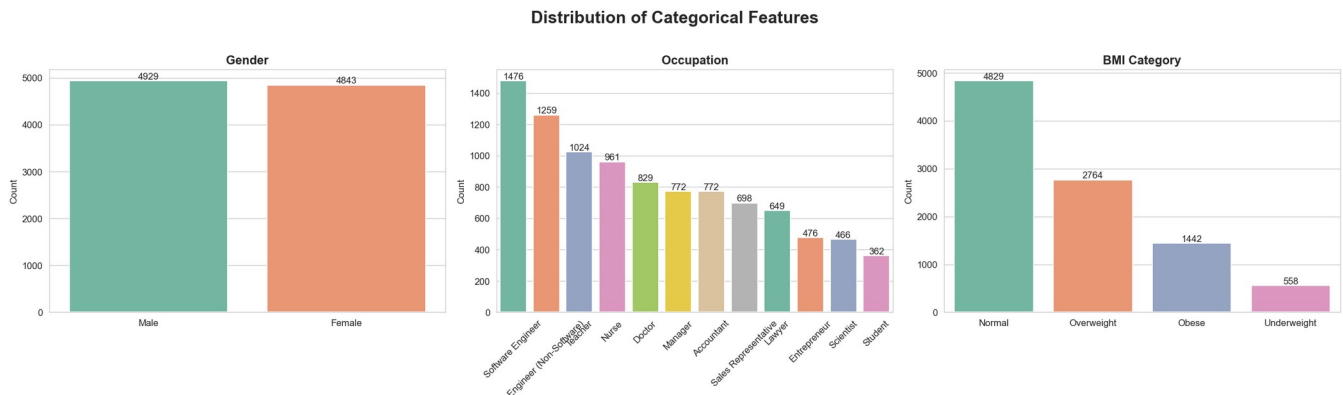


Figure 4.3 — Distribution of Categorical Features

Insights: Gender is close to evenly split, with Male (4,929) and Female (4,843) observations. Occupation spans 13 categories, with Software Engineer (1,476) and Lawyer (1,259) among the most represented groups, while Student (362) has the lowest representation. Just under half of individuals fall in the Normal BMI category (4,829), followed by Overweight (2,764) and Obese (1,442), while Underweight (558) is comparatively less represented.

4.4 Feature Relationships with the Target

Each numerical feature was compared across the three target classes using box plots to identify which signals separate healthy individuals from those with a disorder.

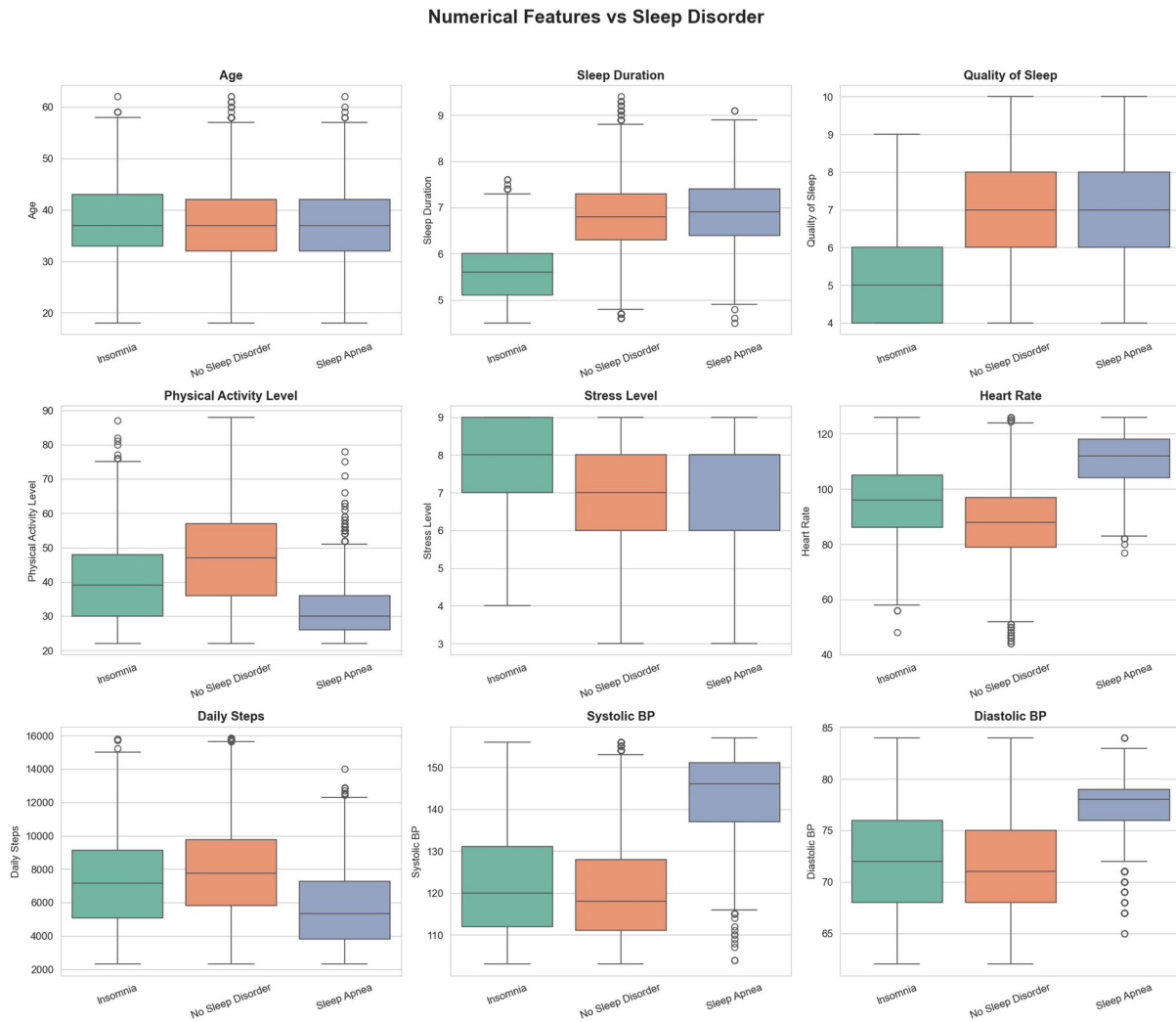


Figure 4.4a — Numerical Features vs. Sleep Disorder

Insights: Sleep Apnea patients show markedly higher Systolic BP, Diastolic BP, and Heart Rate than the other two classes, alongside lower Physical Activity Level and Daily Steps. Insomnia is strongly associated with shorter Sleep Duration, lower Quality of Sleep, and elevated Stress Level. Age distributions largely overlap across all three classes, offering limited standalone separation.

The same comparison was repeated for the three categorical features to test for demographic or occupational association with the target.

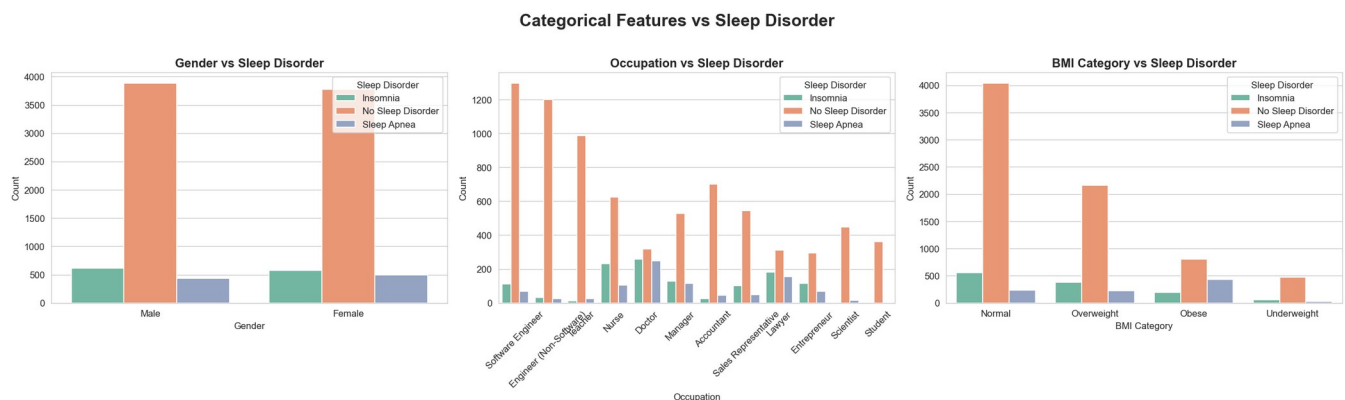


Figure 4.4b — Categorical Features vs. Sleep Disorder

Insights: No Sleep Disorder dominates both genders with similar class distributions, indicating limited target association for Gender. Class distributions vary noticeably across Occupation, suggesting it may provide useful predictive information. BMI Category shows the clearest visual association — No Sleep Disorder dominates the Normal BMI group, while Sleep Apnea is relatively more prominent among Obese individuals.

4.5 Correlation Analysis

A Pearson correlation matrix of the numerical features was computed to surface linear relationships and flag potential multicollinearity ahead of model training.

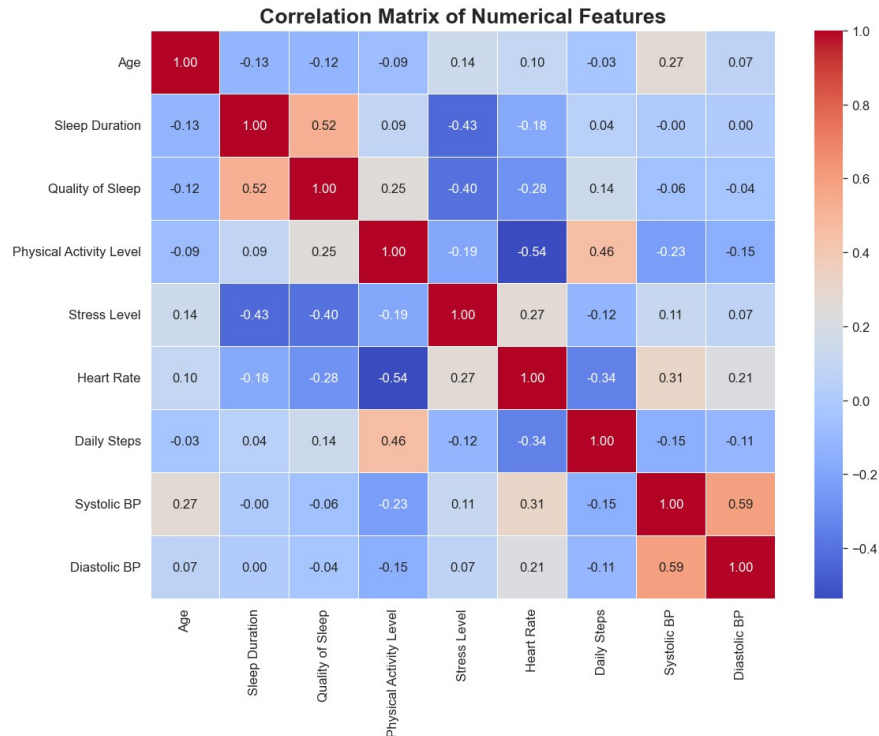


Figure 4.5 — Correlation Matrix of Numerical Features

Insights: Sleep Duration and Quality of Sleep show a moderate positive correlation ($r = 0.52$), as expected for two related sleep metrics. Sleep Duration is also moderately negatively correlated with Stress Level ($r = -0.43$), and Quality of Sleep follows a similar pattern ($r = -0.40$). Physical Activity Level and Heart Rate are moderately negatively correlated ($r = -0.54$), while Physical Activity Level and Daily Steps are moderately positively correlated ($r = 0.46$). Systolic BP and Diastolic BP are moderately-to-strongly positively correlated ($r = 0.59$), confirming these are related but non-redundant cardiovascular signals. No pair reaches the near-perfect correlation that would force feature removal, so all numerical predictors were retained for modelling.

4.6 Multivariate Relationship Analysis

A pair-plot of the most target-relevant numerical features (Sleep Duration, Quality of Sleep, Stress Level, Heart Rate) was generated, coloured by class, to visually inspect how well these signals separate the three sleep-disorder categories in combination.

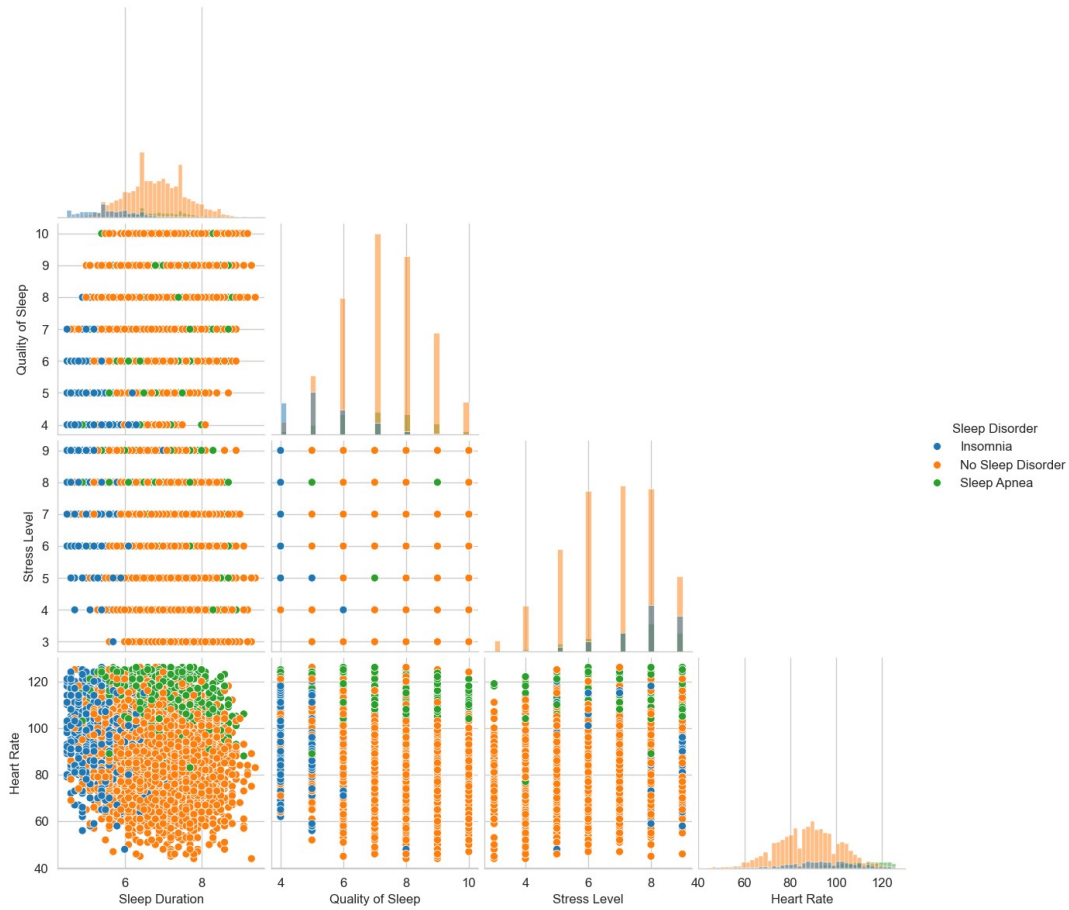


Figure 4.6 — Pair Plot of Key Features by Sleep Disorder Class

Insights: Insomnia (blue) clusters toward shorter Sleep Duration, lower Quality of Sleep, and higher Stress Level, while Sleep Apnea (green) spreads toward higher Heart Rate values, and No Sleep Disorder (orange) dominates the moderate-stress, moderate-heart-rate region — evidence that even simple, partially-separable structure exists in the feature space ahead of formal modelling.

4.7 Outlier Analysis

Outliers in the nine original numerical features were assessed using the Interquartile Range (IQR) method and visualised with box plots.

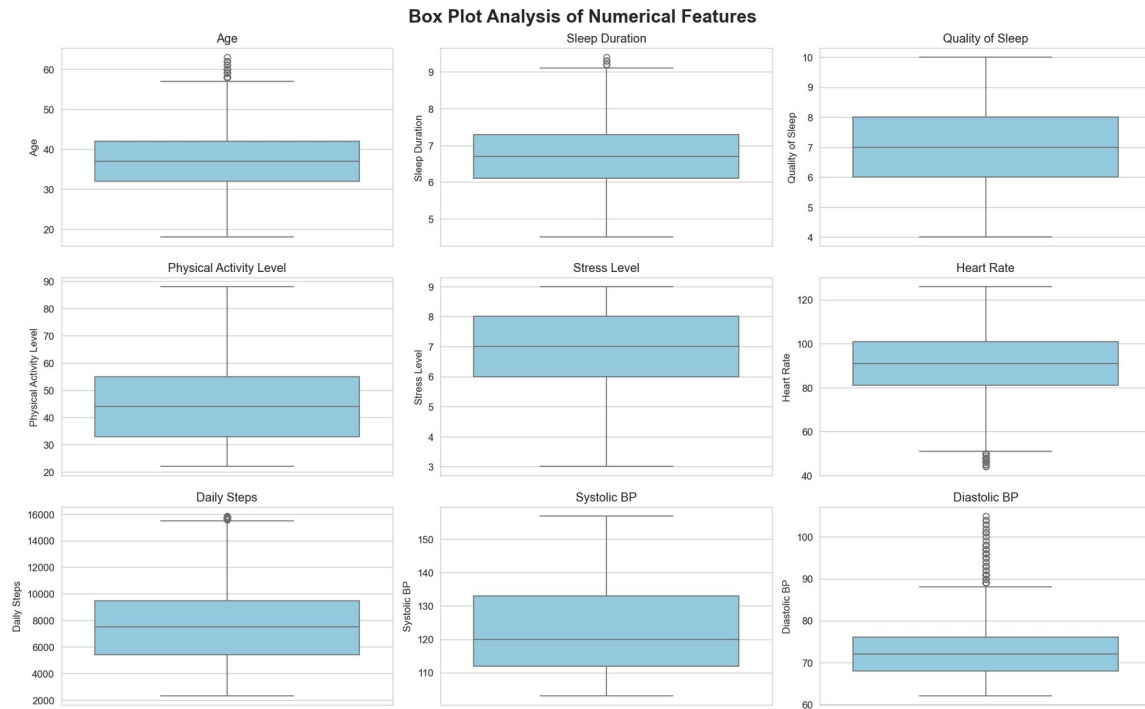


Figure 4.7 — Box Plot Analysis of Numerical Features (Outlier Detection)

Key Insights: Diastolic BP shows the highest proportion of statistical outliers at 0.72% of observations, followed by Age (0.41%), Heart Rate (0.27%), Daily Steps (0.14%), and Sleep Duration (0.07%); the remaining numerical features show no IQR-flagged outliers. These points correspond to physiologically plausible individuals (e.g. very low resting heart rate, unusually high blood pressure) rather than data-entry errors, so they were retained rather than removed — consistent with the objective of modelling real-world variation, including at-risk edge cases.

EDA Summary — Target is imbalanced (78% / 12% / 10%) → Macro F1 and class weighting/SMOTE required. Sleep Apnea is driven by elevated blood pressure and heart rate; Insomnia by lower sleep quality, shorter duration, and higher stress. BMI Category shows the clearest categorical association with the target; Gender carries little predictive value. No feature pair is collinear enough to warrant removal → all predictors retained. Outliers are limited to genuine physiological variation (under 1% of rows per feature) and were kept, not removed.

5. Feature Engineering

Two domain-informed features were derived from existing predictors to give the models additional, clinically meaningful signal without introducing any dependency on the target variable. One of these, the Sleep Quality Index, is a custom engineered feature designed specifically for this project rather than a standardised clinical measure, and is described in full below.

5.1 Engineered Features

Feature	Formula	Rationale
Pulse Pressure	Systolic BP – Diastolic BP	Additional cardiovascular signal beyond raw BP

Feature	Formula	Rationale
Sleep Quality Index	$((\text{Sleep Duration} / 24) + (\text{Quality of Sleep} / 10)) / 2$	Consolidated, standardised sleep-profile score

5.2 Sleep Quality Index — A Custom Engineered Feature

The Sleep Quality Index is a custom engineered feature, created specifically for this project by combining Sleep Duration and Quality of Sleep into a single, normalised indicator. It is not an original or standardised clinical measure; rather, it was designed to provide an interpretable, single-score summary of sleep quantity and perceived sleep quality, using the formula shown in Section 5.1 above.

Higher values of the index indicate a relatively stronger overall sleep profile, combining adequate sleep duration with higher perceived sleep quality, while lower values point toward shorter sleep duration, lower perceived quality, or both. As shown in Section 5.3 below, the index separates the target classes meaningfully and was retained for modelling alongside the original Sleep Duration and Quality of Sleep features.

5.3 Engineered Feature Validation

Each engineered feature was checked for missing values (none were introduced) and validated by comparing its group-wise mean across the three target classes.

- **Pulse Pressure** shows strong separation for Sleep Apnea — mean 65.61 vs. 50.05 for Insomnia and 48.97 for No Sleep Disorder — directly reflecting the elevated blood pressure observed during EDA.
- **Sleep Quality Index** is lowest for Insomnia (0.378) compared to 0.505 for No Sleep Disorder and 0.495 for Sleep Apnea, consolidating two related sleep metrics into a single, standardised signal.

Both engineered features separate the target classes meaningfully and were retained for modelling.

6. Data Preprocessing & Train-Test Split

6.1 Preprocessing Pipeline

All preprocessing was implemented as a single scikit-learn `ColumnTransformer`, fit exclusively on the training fold in every cross-validation split to prevent information leakage from the test set.

Feature Type	Pipeline
Numerical (11 features)	Median imputation → StandardScaler
Categorical (3 features)	Mode imputation → OneHotEncoder

The numerical group includes the nine original physiological/lifestyle measures plus the two engineered features (Pulse Pressure, Sleep Quality Index). Combining the numerical and one-hot-encoded categorical outputs expands the processed feature matrix from 13 raw columns to **29 model-ready features**.

6.2 Stratified Train-Test Split

Given the 78% / 12% / 10% class imbalance, a random split risks under-representing Insomnia or Sleep Apnea in either partition. A stratified 80/20 split was therefore used to preserve the original class proportions in both the training and test sets.

Split	Samples	Class Proportions Preserved
Training Set	8,000	Yes (stratified) — 6,250 / 990 / 760
Test Set	2,000	Yes (stratified) — 1,563 / 247 / 190

The test set remained completely untouched until the final evaluation stage in Section 8.

7. Model Development & Selection

7.1 Modeling Approach

Five classification algorithms — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost — were each benchmarked under two imbalance-handling strategies, class weighting and SMOTE (Synthetic Minority Over-sampling), using Stratified 5-Fold Cross-Validation. Preprocessing and SMOTE (where applicable) were refit independently inside every fold to guarantee a leakage-safe comparison.

Primary metric: Macro F1, which weights every class equally regardless of size — the appropriate choice for a 78/12/10 imbalanced target. Balanced Accuracy, Macro Precision/Recall, and ROC-AUC were tracked as supporting evidence.

7.2 Baseline Cross-Validation Results

Class-Weighting Strategy

Model	Accuracy	Balanced Acc.	F1 Macro	ROC-AUC
XGBoost	89.96%	76.25%	78.81%	95.17%
Random Forest	88.24%	70.60%	75.95%	—
Logistic Regression	83.31%	85.09%	75.33%	—
SVM	83.38%	83.73%	75.06%	—
Decision Tree	82.59%	68.45%	68.42%	—

SMOTE Strategy

Model	Accuracy	Balanced Acc.	F1 Macro
Random Forest + SMOTE	88.55%	78.04%	78.82%
XGBoost + SMOTE	88.40%	77.15%	78.29%
SVM + SMOTE	85.28%	82.33%	76.65%
Logistic Regression + SMOTE	84.36%	84.81%	76.43%
Decision Tree + SMOTE	81.65%	69.77%	67.89%

XGBoost led the class-weighting strategy outright (78.81% Macro F1), while Random Forest and XGBoost were near-identical under SMOTE (78.82% vs. 78.29%). XGBoost — alongside Random Forest, its closest competitor — was carried forward into hyperparameter tuning.

7.3 Class Weight vs. SMOTE Comparison

The two imbalance-handling strategies were compared directly across all five candidate models on cross-validated Macro F1.

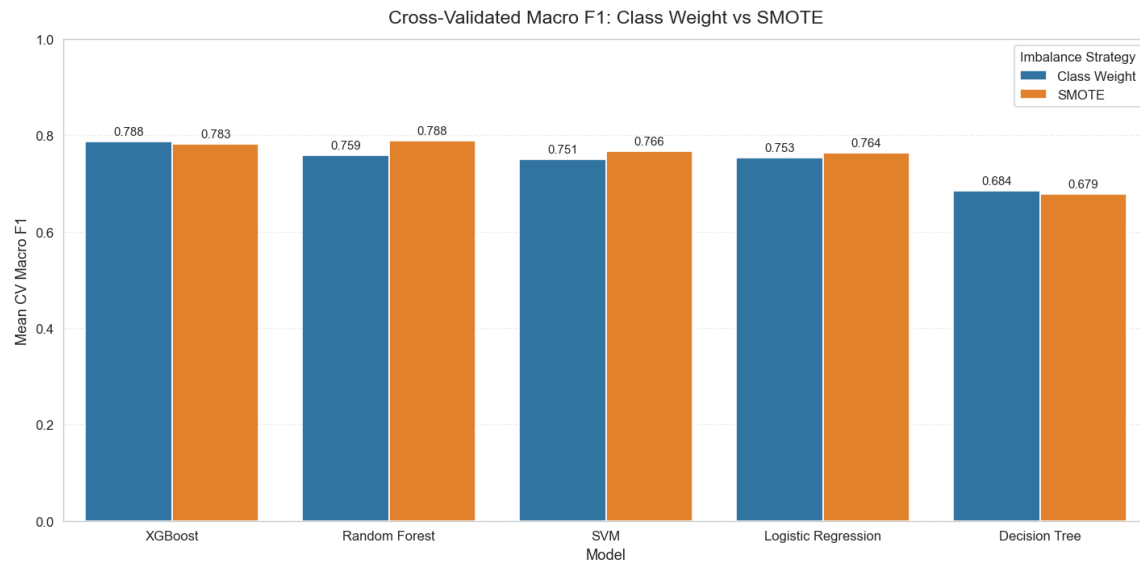


Figure 7.3 — Cross-Validated Macro F1: Class Weight vs. SMOTE

Observation: SMOTE holds a consistent edge for Random Forest, SVM, and Logistic Regression, while class weighting is marginally stronger for XGBoost and Decision Tree. Given the near-parity in performance and XGBoost's lead under class weighting, class weighting was preferred going forward — it avoids generating synthetic samples and is simpler to reproduce consistently at inference time.

7.4 Hyperparameter Tuning

The strongest candidates from the baseline comparison — XGBoost (both strategies) and Random Forest — were tuned using `RandomizedSearchCV` with Stratified 5-Fold cross-validation, optimising for Macro F1. The test set remained untouched throughout this stage.

Candidate	Best CV Macro F1
XGBoost — Class Weight (selected)	80.41%
XGBoost — SMOTE	80.01%
Random Forest — Tuned	78.33%

Selected Configuration — XGBoost (Class Weight)

Hyperparameter	Tuned Value
n_estimators	200
max_depth	3
learning_rate	0.10
subsample	0.70
colsample_bytree	1.0
min_child_weight	1

SELECTED — XGBoost (Class Weight, tuned) selected as the final model — highest cross-validated Macro F1 (80.41%) among all candidates tested.

8. Final Model Evaluation

The tuned XGBoost model was evaluated once, on the 2,000-record held-out test set that had not been used during training, tuning, or cross-validation — giving a realistic estimate of production performance.

8.1 Held-Out Test Set Performance

Metric	Score
Accuracy	90.95%
Balanced Accuracy	81.38%
F1 Macro	83.41%
F1 Weighted	90.74%
Precision Macro	85.78%
Recall Macro	81.38%
ROC-AUC (One-vs-Rest)	96.77%
PR-AUC (Macro Avg., One-vs-Rest)	90.02%

The test Macro F1 (83.41%) exceeds the cross-validated average (80.41%), and overall accuracy (90.95%) comfortably clears the 78% no-skill baseline of always predicting the majority class — evidence of genuine generalisation from cross-validation to an unseen test set.

8.2 Confusion Matrix

The confusion matrix breaks down prediction outcomes by true class to reveal exactly where the model succeeds and where it confuses classes.

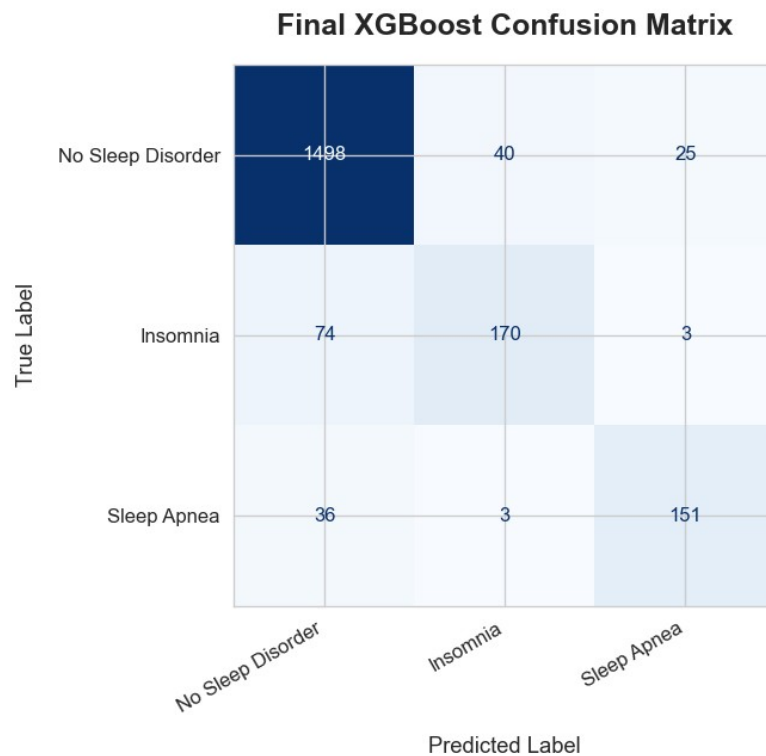


Figure 8.2 — Final XGBoost Confusion Matrix (Test Set)

Key Insights: the model correctly classifies 1,498 of 1,563 No Sleep Disorder cases and 151 of 190 Sleep Apnea cases. Insomnia is the hardest class to isolate — 74 of 247 true Insomnia cases are misclassified as No Sleep Disorder, while confusion between Insomnia and Sleep Apnea is minimal (at most 3 records each direction), showing the model rarely confuses the two disorder types with each other — only with the healthy baseline.

8.3 Multiclass ROC-AUC

One-vs-Rest ROC curves were plotted for each class to evaluate ranking quality independent of the classification threshold.

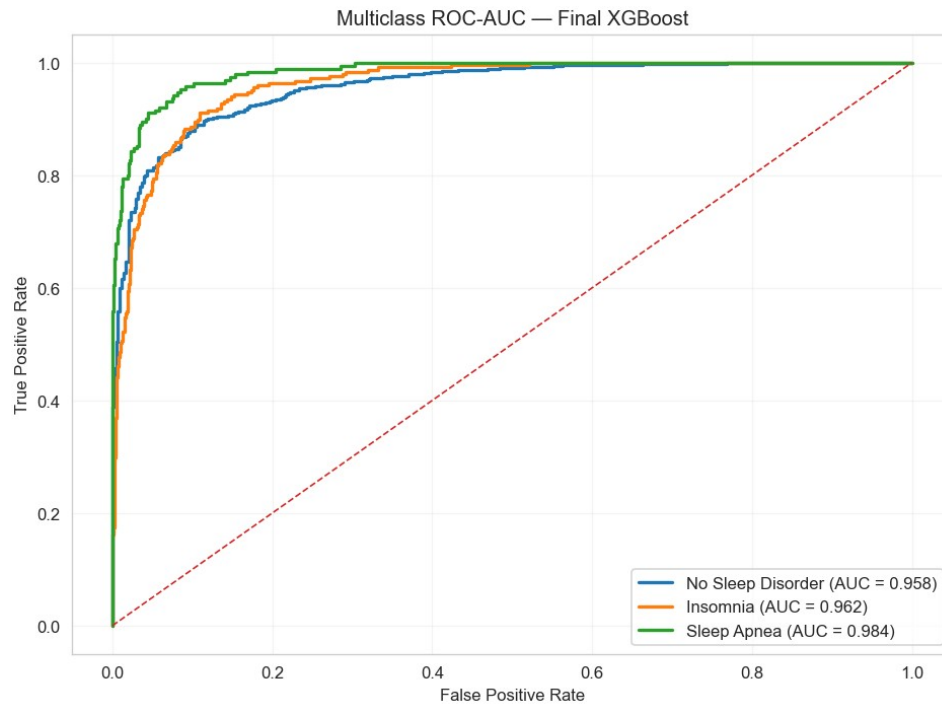


Figure 8.3 — Multiclass ROC-AUC Curve, Final XGBoost Model

Insights: all three classes achieve AUC at or above 0.958 — No Sleep Disorder 0.958, Insomnia 0.962, and Sleep Apnea 0.984 — indicating strong discrimination across every decision threshold, not just the default 50% cut-off used for the confusion matrix above.

8.4 Precision-Recall Curves & PR-AUC

Because the target is imbalanced, Precision-Recall analysis was used alongside ROC-AUC to give a second, threshold-independent view of class-specific performance. One-vs-Rest PR curves were plotted for each class using the final XGBoost model, with particular attention to the more difficult, less-separated classes.

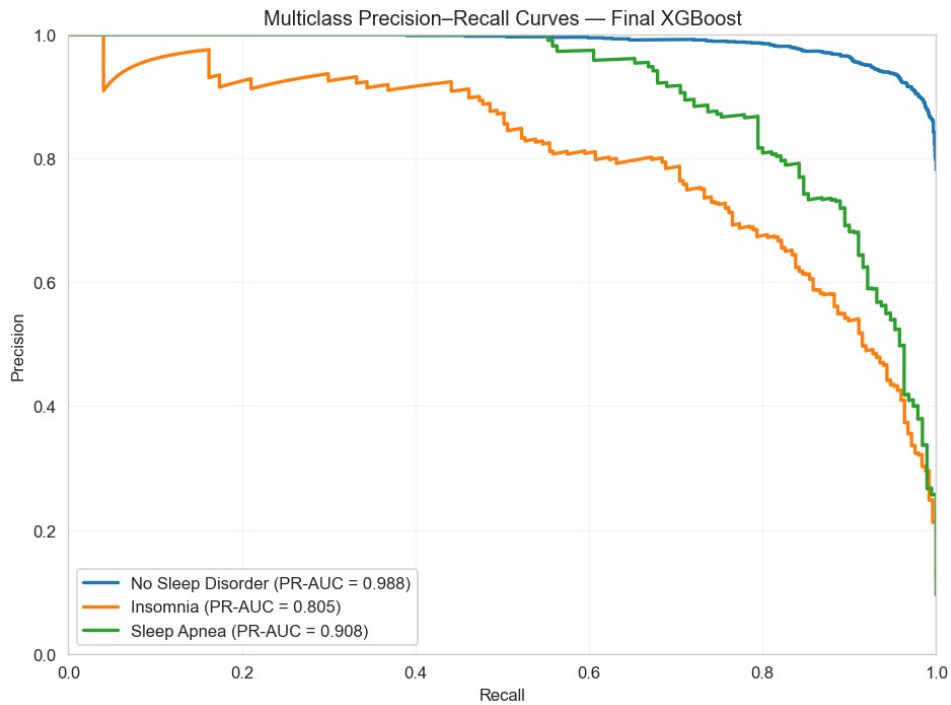


Figure 8.4 — Multiclass Precision-Recall Curves, Final XGBoost Model

Insights: PR-AUC is 0.988 for No Sleep Disorder and 0.908 for Sleep Apnea — both comfortably above 0.90 — while Insomnia trails at 0.805. Because PR-AUC is more sensitive than ROC-AUC to performance on the minority classes, this gap reinforces the confusion-matrix finding that Insomnia is the model's hardest class to separate from a healthy profile, even though its ROC-AUC alone (0.962) looks strong.

8.5 Class-wise Performance

Precision, Recall, and F1-score were computed independently for each class to identify which disorder is hardest for the model to detect.

Class	Precision	Recall	F1-Score
No Sleep Disorder	93.16%	95.84%	94.48%
Sleep Apnea	84.36%	79.47%	81.84%
Insomnia	79.81%	68.83%	73.91%

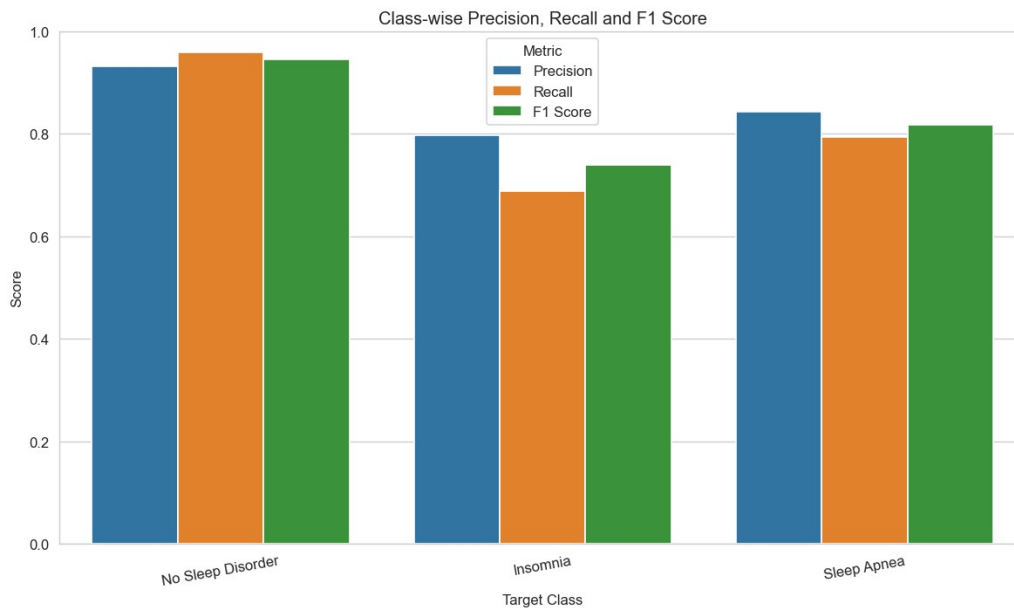


Figure 8.5 — Class-wise Precision, Recall, and F1-Score

Reading the results: No Sleep Disorder is detected very reliably ($F1 = 94.48\%$), Sleep Apnea remains solidly detected ($F1 = 81.84\%$), while Insomnia is the hardest class ($F1 = 73.91\%$) and is most often confused with a healthy profile — consistent with the EDA finding that Insomnia's physiological signals overlap more with No Sleep Disorder than Sleep Apnea's more distinctive cardiovascular profile does.

9. Probability Calibration Analysis

Beyond raw classification accuracy, the reliability of the final XGBoost model's predicted probabilities was examined as a separate post-training validation experiment. Probability calibration evaluates whether the predicted probabilities correspond to the observed frequency of each sleep-disorder class — a well-calibrated classifier assigns probabilities that reflect the actual likelihood of the corresponding outcome. Calibration is evaluated separately for No Sleep Disorder, Insomnia, and Sleep Apnea using a one-vs-rest approach, combining calibration curves, quantitative probability metrics, and a cross-validated comparison between sigmoid-calibrated (Platt Scaling) and isotonic-calibrated versions of the model.

9.1 Probability Calibration Curves

Calibration curves compare the mean predicted probability within ten uniform probability bins against the observed frequency of each class, evaluated separately for No Sleep Disorder, Insomnia, and Sleep Apnea using a one-vs-rest approach. A perfectly calibrated model would fall exactly on the diagonal reference line.

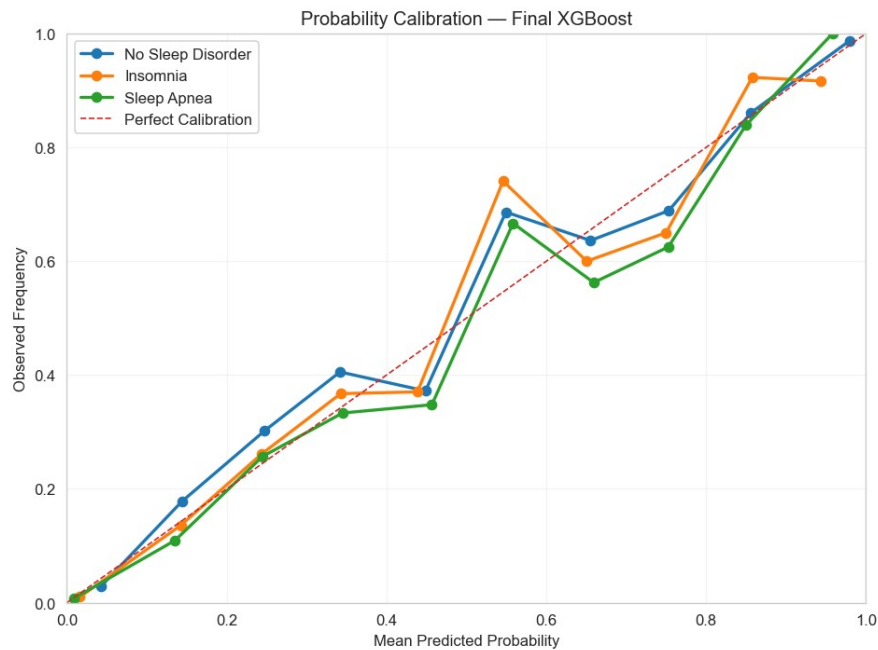


Figure 9.1 — Probability Calibration Curves, Final XGBoost Model

The calibration curves compare the predicted probabilities with the observed class frequencies for No Sleep Disorder, Insomnia, and Sleep Apnea. The curves generally follow the perfect-calibration reference line, although some deviation is visible across intermediate probability ranges, particularly for Sleep Apnea.

The results suggest that the original XGBoost probabilities provide a reasonable level of probability agreement, while also indicating that some calibration error remains across intermediate probability ranges. Quantitative probability metrics are therefore considered alongside the calibration curves.

9.2 Calibration Metrics

To complement the calibration curves, Log Loss and the multiclass Brier Score were calculated on the held-out test set to quantify the quality of the predicted probability distributions. Lower values indicate better probability performance.

Metric	Score
Multiclass Log Loss	0.2386
Multiclass Brier Score	0.1366

The final XGBoost model achieved a multiclass Log Loss of 0.2386 and a multiclass Brier Score of 0.1366 on the test set. Combined with the calibration curves, the results indicate that the model's predicted probabilities show a reasonable level of agreement with the observed class frequencies, although some deviation from the perfect calibration line remains, particularly across the mid-probability range.

These results provide evidence that the model probabilities are informative; however, calibration performance should be interpreted alongside the reliability curves and the model's overall classification performance.

9.3 Calibration Method Selection

Platt Scaling (sigmoid) and Isotonic Regression were evaluated as candidate post-training calibration methods for the final XGBoost model. To avoid selecting a calibration method based on its performance on the held-out test set — which

would leak test-set information into the model-selection process — the comparison was instead conducted entirely within the training data, using 5-fold cross-validation. For each method, a `CalibratedClassifierCV` wrapper was cross-validated and the out-of-fold (OOF) Log Loss was recorded; the method with the lower OOF Log Loss was selected for a single, final evaluation on the untouched test set.

9.4 Cross-Validated Calibration Comparison

The table below reports the out-of-fold Log Loss for each candidate calibration method, averaged across five cross-validation folds computed entirely on the training set.

Method	OOF Log Loss (Cross-Validated)
Platt Scaling (Sigmoid)	0.2981
Isotonic Regression	0.2729

Isotonic Regression achieved the lower out-of-fold Log Loss and was therefore selected as the calibration method to be evaluated on the held-out test set.

9.5 Final Test-Set Calibration Evaluation

With Isotonic Regression selected via cross-validation, a single calibrated model was fit on the full training set and evaluated once on the held-out test set — the first and only point at which this test data was used for calibration evaluation.

Metric	Score
Multiclass Log Loss	0.2375
Multiclass Brier Score	0.1362

The isotonic-calibrated model achieved a marginally lower Log Loss and Brier Score than the original, uncalibrated XGBoost model on this held-out test set.

9.6 Final Calibration Evaluation & Deployment Decision

The uncalibrated XGBoost model and the cross-validation-selected Isotonic Regression calibration were compared on the same untouched test set using Log Loss and the multiclass Brier Score. Lower values indicate better probability estimation.

Metric	Uncalibrated	Isotonic (Selected)
Multiclass Log Loss	0.2386	0.2375
Multiclass Brier Score	0.1366	0.1362

Calibration Comparison and Deployment Decision — Isotonic Regression was selected over Platt Scaling using 5-fold cross-validation on the training data (OOF Log Loss of 0.2729 vs. 0.2981), avoiding any use of the test set during method selection. On the held-out test set, the selected Isotonic calibration achieved a Log Loss of 0.2375 and a

Brier Score of 0.1362, a marginal improvement over the original, uncalibrated XGBoost model (0.2386 and 0.1366). Since this improvement is small, the original, uncalibrated XGBoost model is retained for deployment for simplicity and consistency. The calibration experiments are documented as post-training validation analyses, while the production prediction pipeline remains unchanged.

10. Model Explainability (XAI)

To ensure the final XGBoost model is not a 'black box', SHAP (SHapley Additive exPlanations) was applied via `TreeExplainer` on the transformed test set to quantify how much each feature contributes to every prediction, both globally and by feature interaction.

10.1 SHAP Global Feature Importance

Aggregating SHAP values across all test observations, split by target class, produces a global importance ranking for every feature in the model.

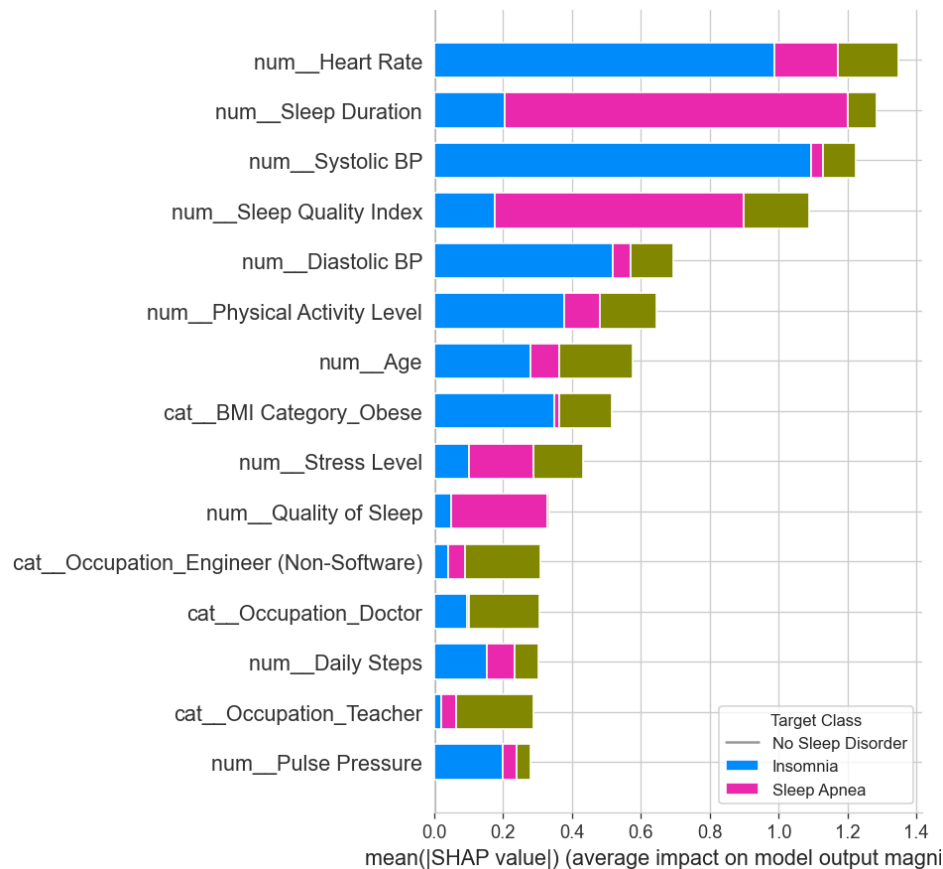


Figure 10.1 — SHAP Global Feature Importance by Target Class

Insights: Heart Rate is the single strongest predictor across all classes, followed by Sleep Duration and Systolic BP. The engineered Sleep Quality Index and Diastolic BP contribute substantially, while Physical Activity Level and Age provide meaningful supporting signal. BMI Category (Obese) is the strongest categorical contributor, ahead of Stress Level and Quality of Sleep. Demographic and occupational one-hot features (e.g. Occupation_Doctor, Occupation_Teacher) and the engineered Pulse Pressure feature rank lowest overall — the model relies primarily on physiological and sleep-related signals rather than demographic proxies.

10.2 SHAP Feature Interaction Analysis

An interaction summary was generated for the model's top continuous predictors to examine whether pairs of features jointly influence predictions beyond their individual, additive effects.

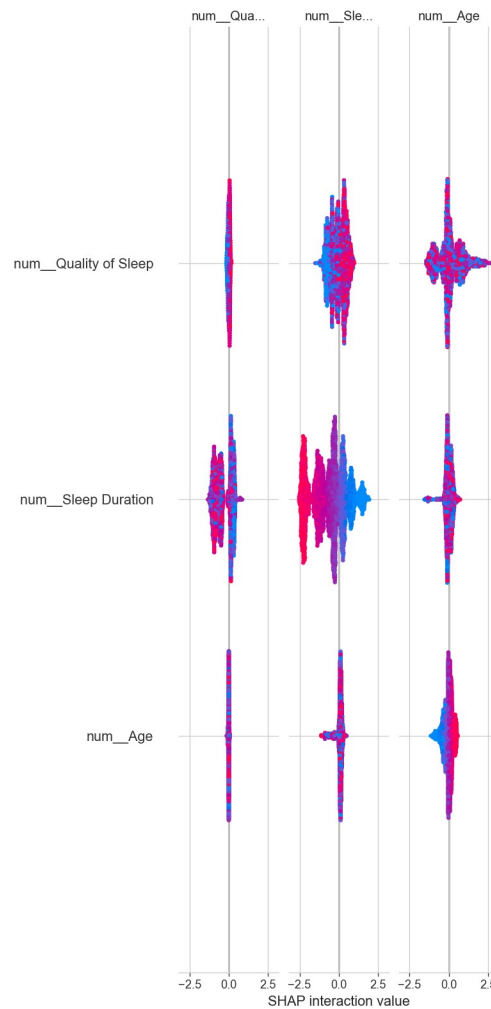


Figure 10.2 — SHAP Feature Interaction Summary (Quality of Sleep, Sleep Duration, Age)

Key Insight: Sleep Duration shows the strongest interaction effect with Quality of Sleep — the impact of one depends visibly on the level of the other, reflecting their real-world relationship as two facets of the same underlying sleep-health construct rather than fully independent signals.

10.3 Explainability Summary

Key Insight — Heart Rate, Sleep Duration, and Systolic BP dominate the model's predictions across both the global SHAP ranking and the interaction analysis. The engineered Sleep Quality Index provides strong secondary signal alongside Diastolic BP, while demographic and occupational features contribute comparatively little. This consistent, human-readable explanation is exactly what makes the model suitable for a real-world screening context, where 'why' matters as much as 'what'.

11. Conclusion & Key Takeaways

11.1 Summary of Findings

- **XGBoost** achieved the strongest and most consistent performance among all five algorithms tested, under both class-weighting and SMOTE imbalance strategies.
- **Class weighting and SMOTE performed comparably** across most models, with SMOTE giving Random Forest, SVM, and Logistic Regression a modest edge — class weighting was preferred for the final model for its simplicity at inference time.
- **Heart Rate, Sleep Duration, and Systolic BP** are the strongest predictors of sleep-disorder status, confirmed independently by both the correlation analysis and SHAP-based explainability.
- **Sleep Apnea** is driven primarily by elevated blood pressure and heart rate; **Insomnia** by lower sleep quality and duration combined with higher stress.
- **Insomnia** remains the hardest class to separate from a healthy sleep profile — the clearest target for future feature or data improvements.

11.2 Final Recommendation

The tuned **XGBoost (Class Weight)** model is recommended as the deployment candidate for this classification task, combining solid held-out test performance (**90.95% accuracy, 83.41% Macro F1, 96.77% ROC-AUC**) with full interpretability via SHAP — a combination essential for any model intended to support health-related screening decisions.

11.3 Business Impact & Recommendations

- Position the model as a **screening / triage aid**, not a diagnostic system — flagged cases should be routed to a qualified healthcare provider for confirmation.
- **Prioritise Insomnia recall** in future iterations, given it is the class most often confused with a healthy profile and therefore the highest-risk false negative.
- Use the SHAP-based explanations to give end users (clinicians or individuals) a transparent, feature-level rationale alongside every prediction, not just a class label.
- Validate the pipeline on real clinical data before any production use, since the current dataset is synthetically generated.

11.4 Skills & Techniques Demonstrated

- Data cleaning, validation, and exploratory data analysis (pandas, seaborn, matplotlib).
- Imbalanced-classification methodology: stratified splitting, class weighting, and SMOTE, compared empirically rather than assumed.
- Feature engineering for health and lifestyle data: pulse pressure and a composite sleep-quality index.
- Leakage-safe pipeline design using scikit-learn's ColumnTransformer and Pipeline, with resampling refit inside every cross-validation fold.
- Model development across linear (Logistic Regression), tree-based (Decision Tree, Random Forest), kernel-based (SVM), and gradient-boosted (XGBoost) paradigms.
- Hyperparameter optimisation using RandomizedSearchCV with Stratified K-Fold cross-validation.
- Explainable AI (XAI) using SHAP TreeExplainer — global feature importance and feature-interaction analysis.
- Clear, structured technical communication — translating model internals into business-relevant conclusions for non-technical stakeholders.

— End of Report —